

Anima: Dimensional LLM Readouts

Technical Overview & Methodology

Version 1.0 (beta) | Siddarth Boggarapu | github.com/Siddarthb07/Anima | MIT License

1. Introduction

Anima is open-source research instrumentation for Hugging Face causal language models. It records internal transformer activations during encoding or generation and maps them to three scalar dimensional readouts: valence (-1 to +1), arousal (0 to 1), and uncertainty (0 to 1). These are constructed measurements from learned linear probes and logits diagnostics - not claims about machine consciousness or clinical affect.

Dimension	Range	Interpretation
Valence	-1 to +1	Pleasant vs unpleasant (Russell circumplex)
Arousal	0 to 1	Activated vs calm
Uncertainty	0 to 1	Epistemic spread (probe + auxiliary signals)

What Anima is

- Measurement harness: forward hooks -> probe heads -> REST/WebSocket API -> dashboard
- Training toolkit: GoEmotions text path or Narratives-style fMRI alignment
- Benchmark runner with manifest JSON for reproducibility

What Anima is not

- Not a chat product, Ollama integration, or feelings detector
- Not clinical or high-stakes decision tooling
- TRIBEv2 pathway is surrogate visualization, not voxel-level fMRI decoding

2. Problem Statement & Design Goals

LLMs produce fluent text, but text alone is an incomplete window into internal computation. Anima separates observed text, hidden activations, and optional brain BOLD signals. It bridges text and activations at inference, and optionally aligns activations with neural data during training.

Principle	Implementation
Honest scope	Random probes work for plumbing; science needs trained checkpoints
Per-token resolution	Hooks on every autoregressive forward pass
Multi-signal uncertainty	Probe + entropy + logit gap + attention entropy
Layer transparency	Softmax fusion; early/late disagreement detection
Reproducibility	CLI, benchmarks, probe zoo metadata sidecars

3. System Architecture

High-level data flow

```
User prompt -> ActivationExtractor (HF causal LM) -> Forward hooks on N layers
  -> AffectProbe (valence/arousal/uncertainty)
  -> Uncertainty from logits + attention
  -> TRIBEv2 surrogate (optional)
  -> Guard + region labeling + suppression
  -> FastAPI (/encode, /generate, WebSocket) -> React dashboard
```

Layer	Tech	Role
Runtime	PyTorch + transformers	Load LM, hooks, encode/generate
Core	core/	Hooks, extraction, guard, suppression
Probes	probes/	AffectProbe, training, zoo I/O
Alignment	alignment/	Narratives, TRIBE surrogate
API	FastAPI	REST + WebSocket on port 8010
Dashboard	Vite + React	Circumplex, uncertainty bars

4. Measurement Methodology

4.1 Activation capture

core/hooks.py registers forward hooks on transformer blocks, capturing residual stream tensors [batch, seq_len, hidden_dim]. encode_sequence() runs one forward pass per input token; extract_iter() streams autoregressive generation with KV cache.

4.2 AffectProbe network

Step 1: softmax-weighted fusion across N probed layers. Step 2: three linear heads - valence (tanh, -1..1), arousal (sigmoid, 0..1), uncertainty (sigmoid, 0..1).

4.3 Auxiliary uncertainty

```
entropy = normalized softmax entropy H/log(V)
logit_gap = 1 / (1 + max(top1 - top2, 0))
attn_entropy = mean head entropy (last layer, final query)
fused = 0.35*entropy + 0.35*logit_gap + 0.30*attn_entropy
```

4.4 Region labeling, suppression, guard

Russell circumplex quadrants via core/regions.py. Suppression flags early vs late layer disagreement (valence_suppression if delta > 0.35). Guard abstains on high fused/probe uncertainty or lexical hedging.

4.5 TRIBEv2 surrogate

Linear map from activations to ROI-like axes (tpj, amygdala, acc, vmppfc, broca). Dashboard visualization only - not neuroscience evidence.

5. Training Pipelines

Path	Command	Summary
Text (GoEmotions)	anima train-text	Emotion labels -> VA via Russell map; MSE loss
Brain (Narratives)	anima train	Word-token align, TR binning, HRF lag, PCA targets
Zoo builder	anima train-zoo	CPU/GPU tiers; download_zoo.py for Release weights

6. API, Dashboard & Evaluation

Endpoint	Purpose
GET /health	Status check
GET /models	Supported models + zoo status
POST /encode	Per-token readouts (single forward)
POST /generate	Autoregressive + suppression
WS /ws/generate	Stream one token message at a time

distilgpt2 benchmark (2026-05-24): Narratives holdout MSE 0.081; Pearson r valence/arousal 0.284/0.004; GoEmotions r 0.057/0.021. Narratives uses synthetic minimal corpus, not full ds002345.

7. Limitations & Ethics

- Uncalibrated by default without probes/zoo checkpoints
- Weak GoEmotions correlation even after training
- TRIBEv2 is a sketch, not brain imaging
- Use readout language; avoid claiming models feel emotions as fact
- Not for clinical, hiring, or law-enforcement use

8. Quick Start

```
git clone https://github.com/Siddarthb07/Anima.git && cd Anima
python scripts/bootstrap.py
anima api --port 8010
cd dashboard && npm install && npm run dev
```